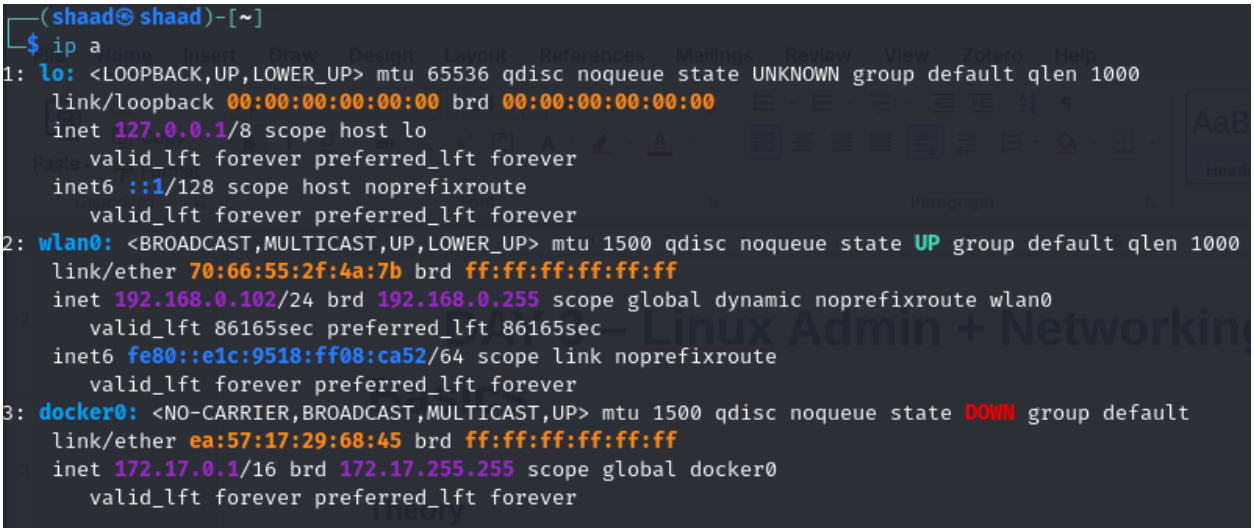
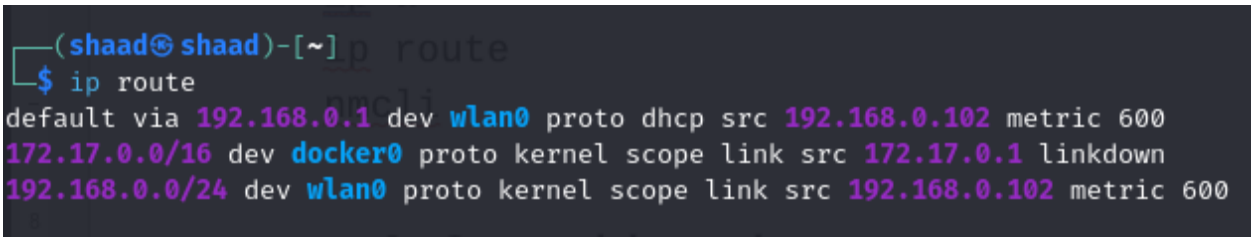
Day – 03 – Linux Admin + Networking basics

Network Configuration

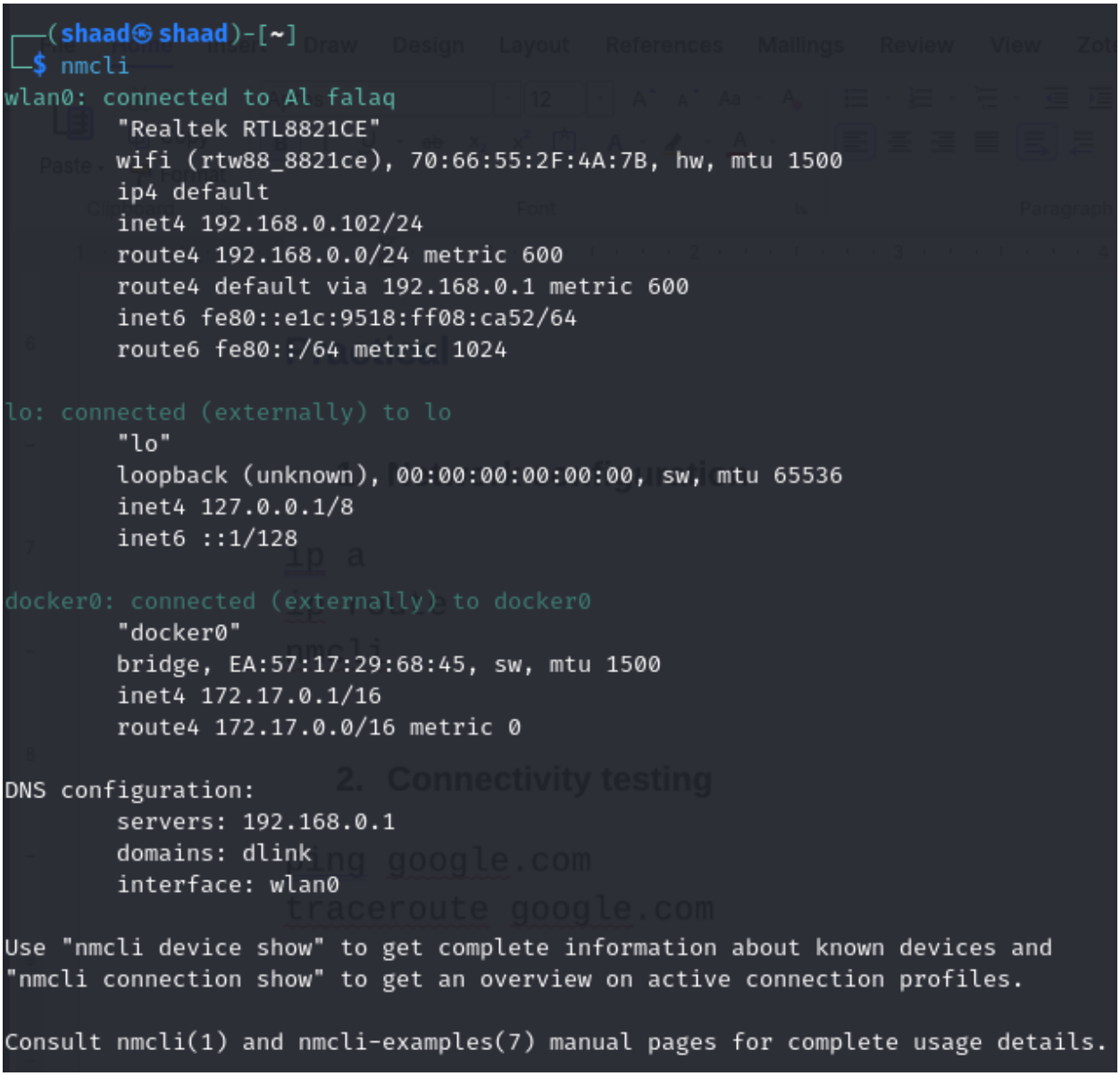
ip a: Shows network interfaces and their IP addresses



ip route : shows systems routing table. Default Gateway and network routes



nmcli : nmcli is a command line tool for controlling NetworkManager and reporting network status



Connectivity testing

Ping google.com : Tests network connectivity by sending packets to Google's server and measuring response time.

```
┌──(shaad㉿shaad)-[~]
└─$ ping google.com -c4
PING google.com (142.251.37.142) 56(84) bytes of data.
64 bytes from pnfjra-aj-in-f14.1e100.net (142.251.37.142): icmp_seq=1 ttl=115 time=24.4 ms
64 bytes from pnfjra-aj-in-f14.1e100.net (142.251.37.142): icmp_seq=2 ttl=115 time=37.4 ms
64 bytes from pnfjra-aj-in-f14.1e100.net (142.251.37.142): icmp_seq=3 ttl=115 time=21.1 ms
64 bytes from pnfjra-aj-in-f14.1e100.net (142.251.37.142): icmp_seq=4 ttl=115 time=35.6 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 21.125/29.632/37.373/6.990 ms
```

traceroute google.com : shows network path (hops) packets take to reach Google's server.

```
┌──(shaad㉿shaad)-[~]
└─$ traceroute google.com
traceroute to google.com (142.251.37.78), 30 hops max, 60 byte packets
 1  _gateway (192.168.0.1)  2.052 ms  2.199 ms  2.161 ms
 2  * * *
 3  10.53.26.193 (10.53.26.193)  101.750 ms  101.717 ms  101.682 ms
 4  10.100.137.26 (10.100.137.26)  101.912 ms  101.877 ms 10.100.136.150 (10.100.136.150)  101.580 ms
 5  10.100.35.86 (10.100.35.86)  204.035 ms  203.999 ms  203.966 ms
 6  * * *
 7  * * *
 8  * * *
```

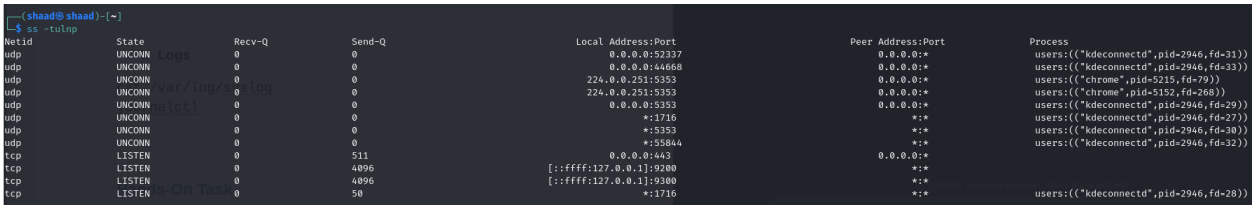
nslookup google.com : Queries DNS servers to find the IP address associated with a domain name

```
┌──(shaad㉿shaad)-[~]
└─$ nslookup google.com
Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.37.142
Name:   google.com
Address: 2a00:1450:4019:819::200e
```

Ports and services

ss -tulnp : shows listening TCP/UDP ports, their associated processes, and numerical addresses



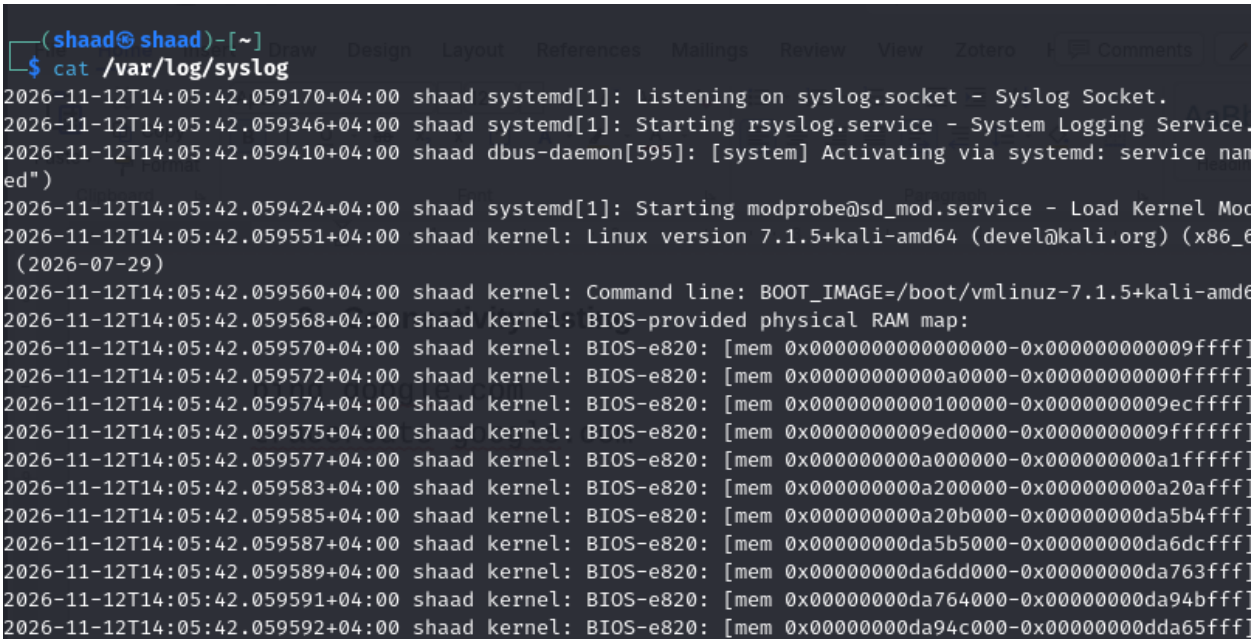
```
(shaad㉿shaad)-[~]
$ ss -tulnp
Netid  State   Recv-Q  Send-Q   Local Address:Port      Peer Address:Port  Process
udp    UNCONN  0       0          0.0.0.0:52337         0.0.0.0:*          users:(("kdeconnectd",pid=2946,fd=31))
udp    UNCONN  0       0          0.0.0.0:44668         0.0.0.0:*          users:(("kdeconnectd",pid=2946,fd=33))
udp    UNCONN  0       0        224.0.0.251:5353        0.0.0.0:*          users:(("chrome",pid=5215,fd=79))
udp    UNCONN  0       0        224.0.0.251:5353        0.0.0.0:*          users:(("chrome",pid=5152,fd=268))
udp    UNCONN  0       0          0.0.0.0:5353          0.0.0.0:*          users:(("kdeconnectd",pid=2946,fd=29))
udp    UNCONN  0       0                *:1716                *:*          users:(("kdeconnectd",pid=2946,fd=27))
udp    UNCONN  0       0                *:5353                *:*          users:(("kdeconnectd",pid=2946,fd=30))
udp    UNCONN  0       0                *:55844               *:*          users:(("kdeconnectd",pid=2946,fd=32))
tcp    LISTEN  0       511        0.0.0.0:443           0.0.0.0:*
tcp    LISTEN  0       4096   [::ffff:127.0.0.1]:9200         *:*
tcp    LISTEN  0       4096   [::ffff:127.0.0.1]:9300         *:*
tcp    LISTEN  0       50               *:1716                *:*          users:(("kdeconnectd",pid=2946,fd=28))
```

netstat -anp : Displays all network connections and listening ports, along with numerical addresses and associated process IDs.

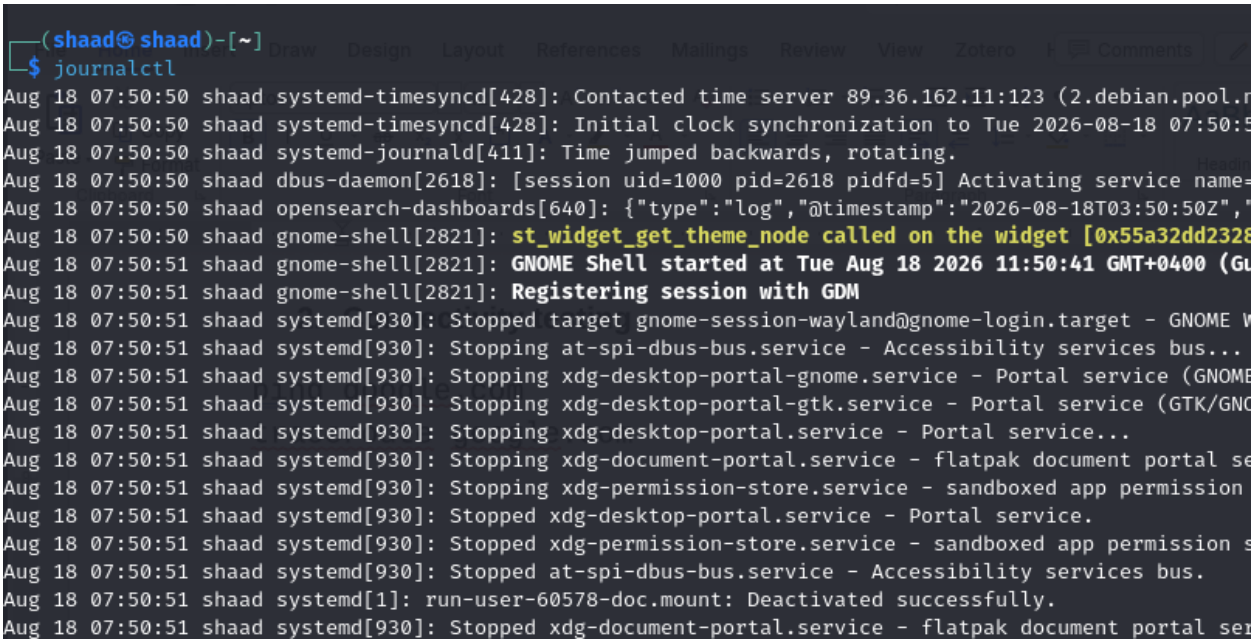
```
(shaad㉿shaad)-[~]
$ netstat -anp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN      -
tcp        0      0 192.168.0.102:39988     173.194.76.188:5228     ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:33688     172.64.155.209:443      ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:33698     172.64.155.209:443      ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:59942     35.190.80.1:443         ESTABLISHED 5215/renderD128 --c
tcp        0      0 127.0.0.1:49388         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:49392         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:49400         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:49410         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 192.168.0.102:60096     104.109.251.217:443     ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:34278     172.64.148.235:443      ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:34290     172.64.148.235:443      ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:34300     172.64.148.235:443      ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:41904     8.232.160.27:443        ESTABLISHED 5215/renderD128 --c
tcp        0      0 192.168.0.102:57648     35.206.118.237:443      ESTABLISHED 5215/renderD128 --c
tcp6       0      0 127.0.0.1:9200          :::*                    LISTEN      -
```

Logs

Cat /var/log/syslog : Displays the system log file containing general system events, messages, and service activity.



Journalctl : Displays system logs collected by systemd-journald.



Hands on Task - Day-03

Identify:

IP address

```
┌──(shaad㉿shaad)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 70:66:55:2f:4a:7b brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.102/24 brd 192.168.0.255 scope global dynamic noprefixroute wlan0
       valid_lft 83090sec preferred_lft 83090sec
    inet6 fe80::e1c:9518:ff08:ca52/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether ea:57:17:29:68:45 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

┌──(shaad㉿shaad)-[~]
└─$
```

Gateway

```
┌──(shaad㉿shaad)-[~]
└─$ ip route
default via 192.168.0.1 dev wlan0 proto dhcp src 192.168.0.102 metric 600
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.0.0/24 dev wlan0 proto kernel scope link src 192.168.0.102 metric 600
```

DNS server

```
┌──(shaad㉿shaad)-[~]
└─$ cat /etc/resolv.conf
# Generated by NetworkManager
search dlink
nameserver 192.168.0.1
```

Explain difference between TCP & UDP

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are two communication protocols used to transfer data over a network.

TCP is a connection oriented protocol that establishes a connection between the sender and receiver before transferring data. It ensures that data packets arrive correctly, in the right order, and without missing information. If packets are lost, TCP retransmits them. It is reliable but generally slower than UDP. Examples include web browsing, email, and file transfers.

UDP is a connectionless protocol that sends data without establishing a connection first. It does not guarantee that packets will arrive, or that they will arrive in the correct order. However, it is faster and has less overhead than TCP. Examples include online gaming, video streaming, and voice calls.